

DATASHEET:

KENYA FOREST SOUNDS DATASET

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This document is based on *Datasheets for Datasets* by Gebru *et al.* [2]. Please see the most updated version [here](#).

MOTIVATION

For what purpose was the dataset created?

This dataset aimed to contribute to existing datasets of sounds collected in forest ecosystems. The data can then be applied to develop technologies for sound classification. It contributes to the state of the art of environmental sound datasets by augmenting natural sounds of tree cutting tools by axe and chainsaw, at the time of project proposal [1].

Who created this dataset?

The Conservation Technology Research Hub at Strathmore University (Kenya) are the primary collectors and owners of the data. It was validated by collaborators from the Kenya Forestry Research Institute (Kenya) and Laboratory of Conservation Biology, University of Neuchâtel (Switzerland).

What support was needed to make this dataset?

The dataset was among the results of a study titled "Curbing Illegal Logging Patterns using Sound-Based Detection Techniques". This project was among the awarded studies in 2023 Climate Change AI Innovation Grants under Grant Number IG-2023-113. Please see the project description [here](#)

COMPOSITION

What do the instances that comprise the dataset represent?

The dataset is comprised of audio clips in a .wav format ranging from 0.07 to 5.02 seconds. Each clip is independent of each other and categorised by a human label.

How many instances are there in total?

This version contains 5500 clips, categorised under the labels identified as in Table 1

Label	Count
Axe	90
Bag	12
Bag Zip	22
Birds	213
car passing	1
Chainsaw	129
Child	2
cow mooing	2
Creaking	1
Crickets	3
Cutting wood	7
Distant talking	15
disturbance noise	6
Drone	5
Engine	3
Flowing water	21
Livestock	26
Machine beeping	1
Machine moving	6
Motorcycle	5
Moving object	8
Moving vehicle	6
Objects moving	46
Objects moving in water	11
passing vehicle	3
people talking	39
people talking and laughing	1
person laughing	2
Person moving	9
Person singing	3
Person talking	3448
person talking over disturbance noise	1
Person walking	379
Phone	4
Plastic wrapper	11
pouring water	4
radio	1
Shuffling	13
sloshing water	4
sloshing water and talking	1
Steel banging	1
Vehicle	703
White Noise	1
Wind	231
SUM	5500

TABLE 1
KFSD LABEL COUNT

Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set?

The total instances collected after preprocessing were 7,531 audio clips. 5,500 clips have been made available in the first version of the dataset to meet time constraints and project publication requirements. These clips were randomly selected during the annotation process, therefore they may not contain all possible instances from the data collection process. The remainder subset will be availed once processed.

What data does each instance consist of?

Each instance contains unprocessed audio.

Is there a label or target associated with each instance?

Yes, the instances are identified with a human label, as listed in table 1.

Is any information missing from individual instances?

There is no missing information from each instance.

Are relationships between individual instances made explicit?

There are no explicit relationships between data instances.

Are there recommended data splits (e.g., training, development/validation, testing)?

During classification, a standard 70/15/15 split between training, testing and validation sets would be appropriate. It should be noted that classes are greatly imbalanced, therefore a label balancing strategy should be applied.

Are there any errors, sources of noise, or redundancies in the dataset?

At the point of publication of the first version of the dataset, blank and unclear audio clips were not included in the uploaded dataset. Redundancies may occur in the labelling of the dataset e.g "Moving vehicle" vs "Car passing". As well, some audio clips feature overlapping sounds where the more dominant sound was provided as the label to the clip.

Is the dataset self-contained, or does it link to or otherwise rely on external resources?

The dataset is maintained by this GitHub repository. Any updates to the repository will be reflected to this endpoint at Zenodo.

Does the dataset contain data that might be considered confidential?

All instances that featured personally identifiable information (PII) data was removed by human reviewers during the annotation process.

Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?

No

Does the dataset relate to people?

No

COLLECTION

How was the data associated with each instance acquired?

The data was directly observable by the researchers at the point of collection. Teams of annotators thereafter independently provided labels to the collected data.

Over what timeframe was the data collected?

Data was collected across three independent instances in March 2024, May 2024, and January 2025. The creation of the data instances (preprocessing) occurred periodically between February 2025 and April 2026.

What mechanisms or procedures were used to collect the data (e.g., hardware apparatus or sensor, manual human curation, software program, software API)?

Data in the field was collected using a Zoom H2n 4-channel Handy Recorder. (See product description here).

What was the resource cost of collecting the data?

The cost of the recording device was estimated at USD 240 at the time of purchase. Other associated costs during data collection supported the researchers during field visits.

If the dataset is a sample from a larger set, what was the sampling strategy?

The approach to data collection followed a simple random sampling approach. The guiding principle was to capture as much natural occurrences during the instances of field visits.

Who was involved in the data collection process and how were they compensated?

Data collection was handled by teams of undergraduate and graduate students, and was coordinated by a project leader. Each student engaged was compensated accordingly as per the project guidelines.

Were any ethical review processes conducted?

Yes, an ethical review was conducted and approved by the Strathmore University Institutional Ethical Review Committee before the data collection process began. Details of the request and approval process can be availed upon request.

Does the dataset relate to people?

Not Applicable.

Any other comments?

The authors acknowledge the efforts by Ali Sheikh, Monique Sankay and Trevor King'ang'i as the team of undergraduate students who supported the data collection process.

PREPROCESSING / CLEANING / LABELLING

Was any preprocessing/cleaning/labelling of the data done?

The dataset went through 2 major stages of preprocessing:

- 1) In the first stage, data was split to 5 second clips using the FFmpeg API in a Python script. The rationale for applying a 5 second split was to match existing standards of related datasets on environmental sounds [1], [3] , and to minimise the computational costs when processing with neural network architectures [4]. The sounds were resampled at a frequency of 16 kHz to generate smaller data sizes to as an effort to avoid computational overhead when applying machine learning classification techniques. The data was then annotated to a specified label. Here, two critical issues emerged: subjective interpretation when labelling sounds and the presence of overlapping sounds. To resolve the challenge of subjectivity, random cross-checking between annotators and reviewers occurred during the annotation process. In the event of overlapping sounds, the clip was labelled by the most prominent sound when reviewed by multiple annotators.
- 2) The second stage largely featured data annotation of unlabelled clips, by human annotators. An observation was the significant presence of human speech in the clips, therefore a transfer learning approach using a fine-tuned Audio Spectrogram Transformer (AST) was applied to sift out instances of human speech in the unlabelled data. This model was applied to 2,505 samples that were manually labelled in the first stage of annotation. Modelling was done on a 70/15/15 split between training, testing and validation and achieved 89.36% accuracy in testing and 92.82% in validation. The model was then applied to a new set of 2,336 unlabelled files, where verification was done by human annotators and 2,334 were correctly labelled as "Person talking" and the remainder were appropriately labelled in other categories. More details including the developed model are available here.

Was the “raw” data saved in addition to the preprocessed/cleaned/labelled data (e.g., to support unanticipated future uses)?

The raw data is available only to the research team, and currently not publicly available. This is to avoid potentially compromising privacy breaches with PII data that could be present in the unannotated subset.

Is the software used to preprocess/clean/label the

instances available?

The FFmpeg API used for initial clipping and frequency resampling is available here. The pre-trained AST is available here while the fine-tuned model used for this study is available here.

Any other comments?

The authors acknowledge the efforts by Lauryn Waruingi, Cicily Muendo, Arnold Ochieng', Shirleen Wanyingi, Michael Barasa, Caldwell Kibe and Lydia Wangaru as the team of undergraduate students who supported the data preprocessing and labelling process.

USES

Has the dataset been used for any tasks already?

No

What (other) tasks could the dataset be used for?

The dataset can be used to detect and analyse human speech due to the significant amount of human speech. A study can be done to process English and Kiswahili speech in use in a natural context

Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?

As presented in table 1, there is a significant class imbalance among identified sound categories. As well, some labels may be conflated to describe similar activities e.g. "Person talking " and "People talking". Strategies to address this imbalance need to be considered e.g. data augmentation.

Are there tasks for which the dataset should not be used?

Data labelled as human speech should not be analysed and applied out of the context of detecting human speech or identifying languages e.g. speech generation and semantic analysis.

DISTRIBUTION

Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created?

No.

How will the dataset will be distributed?

The dataset is distributed through the DOI: **10.5281/zenodo.21443074**

When will the dataset be distributed?

The dataset is currently live and available for distribution through the DOI provided.

Will the dataset be distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)?

The dataset is available under the Creative Commons Attribution 4.0 International license which allows its re-distribution and re-use on the condition that the authors are appropriately credited

Have any third parties imposed IP-based or other restrictions on the data associated with the instances?

No

Do any export controls or other regulatory restrictions apply to the dataset or to individual instances?

No

MAINTENANCE

Who is supporting/hosting/maintaining the dataset?

The Conservation Technology Research Hub at Strathmore University is hosting and maintaining the dataset.

How can the owner/curator/manager of the dataset be contacted (e.g., email address)?

Any questions or concerns can be raised by email to Allan Vikiru at avikiru@strathmore.edu

Is there an erratum?

None at the time of data publication.

Will the dataset be updated (e.g., to correct labeling errors, add new instances, delete instances)?

Yes, updates will be done directly to the public GitHub repository, and be synchronised to the repository on Zenodo.

Will older versions of the dataset continue to be supported/hosted/maintained?

Older versions of the dataset will be available as older releases on the GitHub repository. Any updates will be provided in the description sections of the dataset record on GitHub and Zenodo.

If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so?

Other researchers are free to build on the dataset in any approach that seems fit to their target results, provided suitable citation is provided to the authors. Opportunities to validate or verify are welcome, as requests sent via email through the contact information provided.

REFERENCES

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